Deep-Learning Neural Networks as a Model of Saccadic Generation

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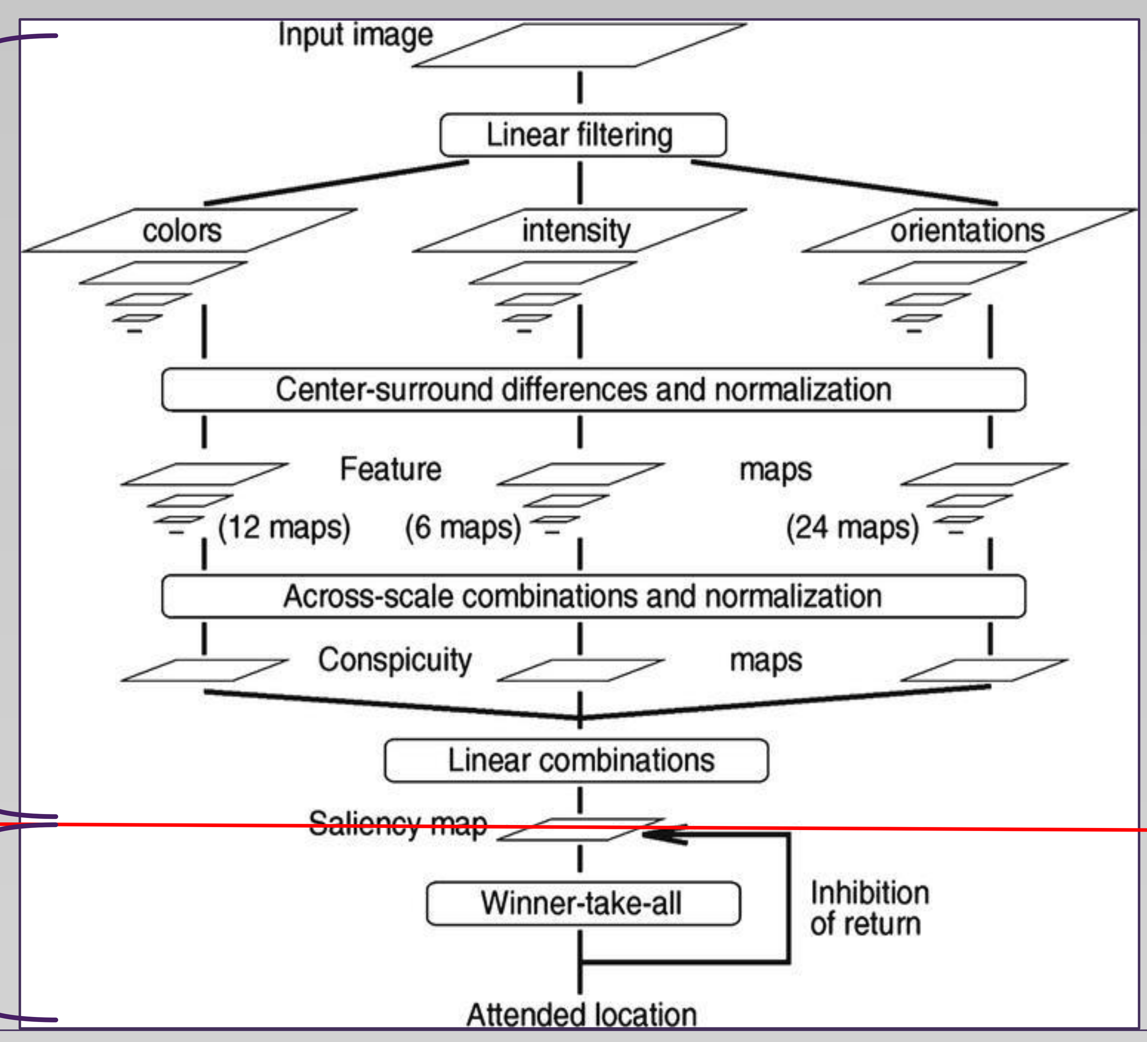
Abstract

Approximately twenty years ago, Laurent Itti and Christof Koch created a model of saliency in visual attention in an attempt to recreate the work of biological pyramidal neurons by mimicking neurons with centre-surround receptive fields. The Saliency Model launched many studies that contributed to the understanding of layers of vision and the sphere of visual attention. The aim of the current study is to improve this model by using an artificial neural network that is able to learn how to generate saccades similar to the way that humans make saccadic eye movements. The proposed model uses a Leaky Integrate-and-Fire layer for temporal predictions, and replaces spatial salience with a deep learning neural network in order to create a generative model that is as biologically precise as possible by combining spatial and temporal contributions of both. The results of the study involve a deep neural network, which is able to predict eye movements based on unsupervised learning from raw image input, as well as supervised learning from fixation maps retrieved during an eye-tracking experiment conducted with 35 participants at later stages in order to train a 2D softmax layer. The results imply that it is possible to match the spatial and temporal distributions of the model to spatial and temporal human distributions.

Spatial module

Temporal module



Laurent Itti & Christof Koch (2000), Model of saliency in visual attention: biological aspect (pyramidal cells in centre-surround receptive fields + feature integration theory (Treisman & Gelade, 1980), attention - multiple object feature recognition and integration. Launched many studies that contributed to the understanding of layers of vision and the sphere of visual attention.

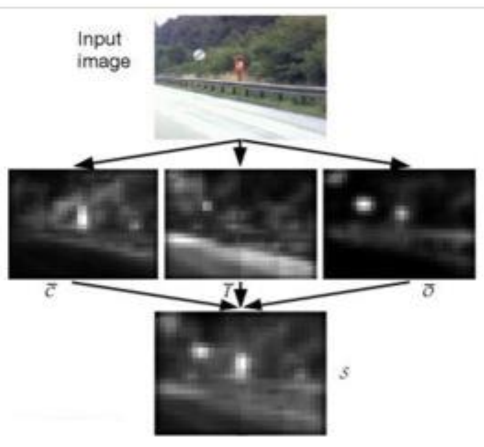
=> Efficient recreation of the visual system: biological accuracy of older models with fewer parameters or complex newer approaches?

Approach Overview: spatial VS temporal

- Interdisciplinary approach - 'reverse engineering'. Computational modelling + human vision research

Spatial

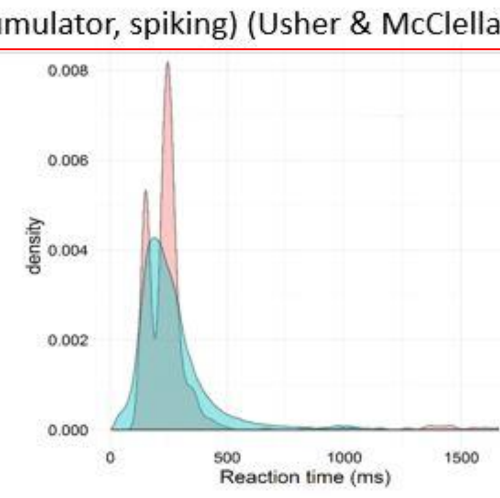
- Itti & Koch – high spatial, low temporal distribution (bio + FIT).
- Visual search paradigm approach: focused on the search efficiency, not SRT (Wolfe, 1997), guided visual search process - visual attention is guided by preceding low-level parallel processes of feature recognition



(Itti, Koch & Niebur, 1998)

Temporal

Not adjusted to visual research independently, require integration into spatial models:
- Diffusion (binary:2 options) (Ratcliff & McKoon, 2008)
- Race (>2 options, WTA) (Purcell, 2012; Wolfe, 2007)
- LIF (accumulator, spiking) (Usher & McClelland, 2001)



(MacInnes, 2017)

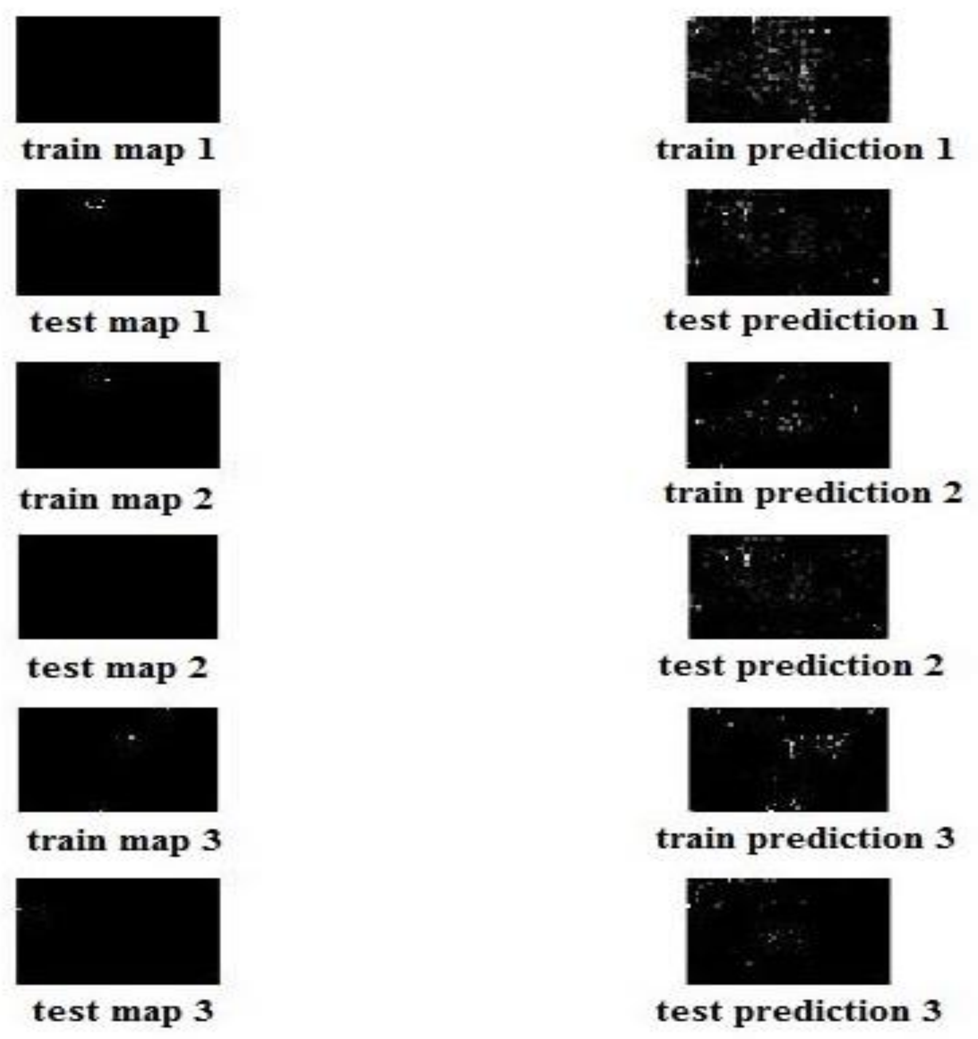
Leaky Integrate-and-Fire

- Similar to a biological neural network.
- 'Spikes': neurons fire upon reaching a threshold.
- 'Leaks' of accumulated data when the input signal ceases.
- Accurate 2D spatial representations of visual space + biological lateral inhibition

Methods

	Model1 (Proof of concept)	Model 2 (Failed)	Model 3
Input	Real images: 10 training, 4 testing	Real images: 10 training, 4 testing + averaged human fixation maps from 35 participants: 10 training, 4 testing	Real images: 59 training, 18 testing (feed forward) + human fixation maps: 1115 training, 478 testing (feedback for softmax) (individual data from 35 participants)
Supervised	No	Yes (early layers remained unsupervised)	Yes (early layers remained unsupervised)
Filters required	Yes	No	No
Toolbox	DeeBNet, Saliency Toolbox	DeeBNet, Saliency Toolbox	Matlab Neural Network Toolbox, Saliency Toolbox
Spatial component algorithm	RBM	RBM + Softmax	Autoencoders (2 layers) + Softmax (1 layer): 1) Autoencoders: 200 iterations, hidden size = 50; 2) Softmax: default parameters at 100 iterations; 3) Stacking of 3 layers; 4) Testing on testing and training data.

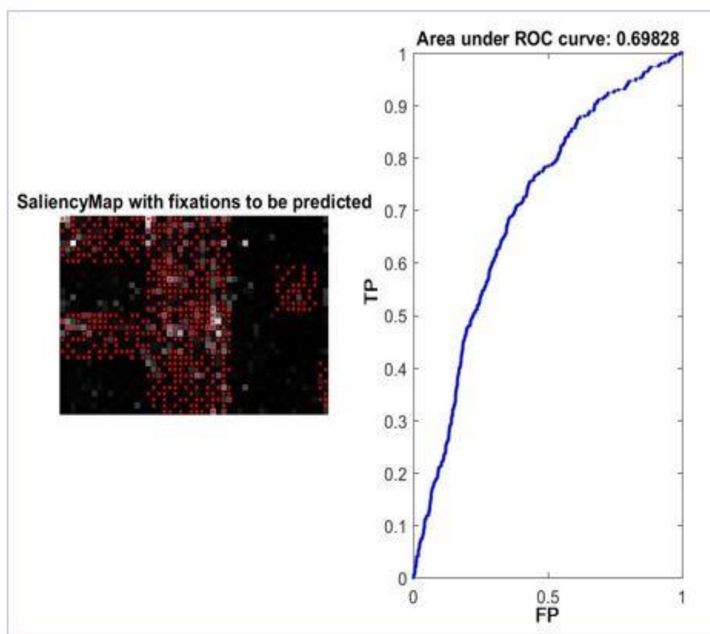
Results: Spatial



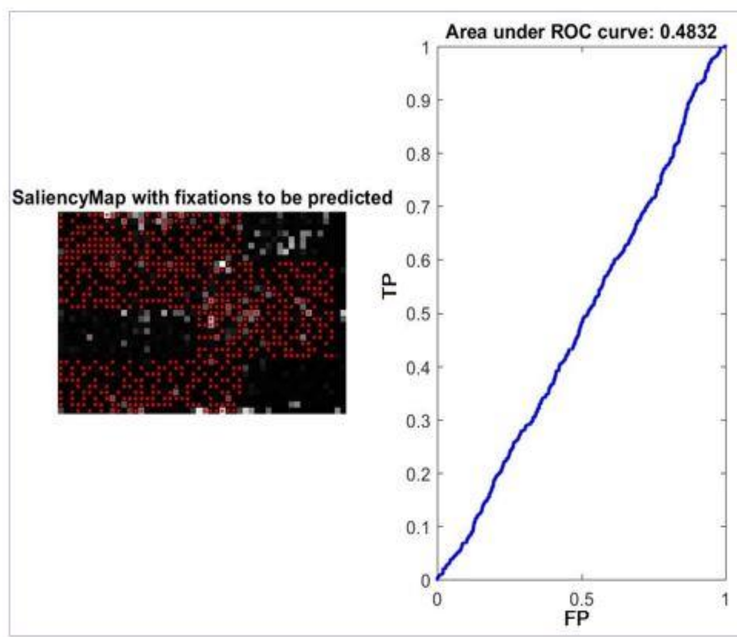
NN Spatial Predictions

AUC-Judd Metrics for Spatial Predictions: Above chance

- The average score on the test for 77 generated maps was estimated as 0.60

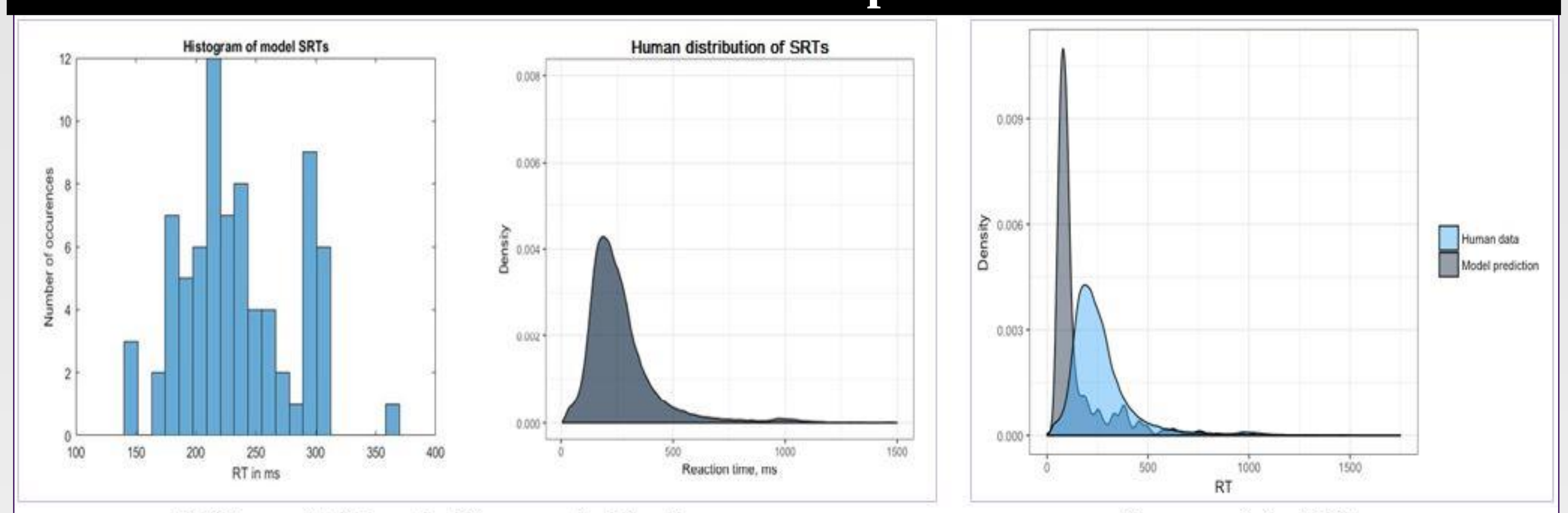


The highest score among the 77 generated maps



The lowest score among the 77 generated maps

Results: Temporal



DLNN model RT against human distribution

Merzon et al., 2018

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